Crab_Habs

Ole Shelton

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Intro

Some words about Habs and Crabs.

Data

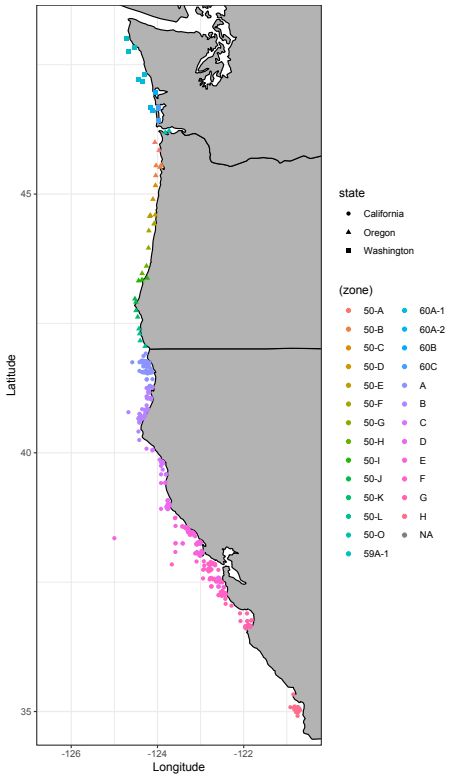


Figure 1: A map of the different location in the habs-crabs dataset.

I filtered the data:

- Only included time-series with:
  - Viscera observations.
  - at least 4 observations within a zone-crab_year combination
  - at least one observation within a zone-crab_year was > 30 ppm domoic acid.

This results in 42 survey time-series. They look like this



Figure 2: Caption. Reference day 0 = September 1. red line = 30 ppm domoic acid

It feels pretty clear that there is a lot of noise in the measurement of domoic acid and that there is a censoring problem so I did this:

- Data were considered censored
  - below 1ppm in Oregon and Washington
  - below 2.5ppm in California

I made a pretty straight forward state-space model for the Dungeness Crab time-series. Crab year starts on September 1 of each year. Model starts on the first day in which there was an observation > 30. I made a stochastic exponential decay model with X_{it} being the true domoic acid concentration at site i on day t. The model is linear in log-space,

$$\log(X_{i,t}) = \log(X_{i,t-1}) + \alpha + \epsilon_{i,t} \tag{1}$$

$$\epsilon_{i,t} \sim N(0, \tau) \tag{2}$$

with α being the decay rate parameter and ϵ being the process variability for each time step (day) with standard deviation τ. In practice, we do have observations each day but surveys occur at irregular intervals. Let δ be the number of days between surveys and then we actually estimate,

$$\log(X_{i,t}) = \log(X_{i,t-\delta_{it}}) + \delta_{it}\alpha + \epsilon_{i,t} \tag{3}$$

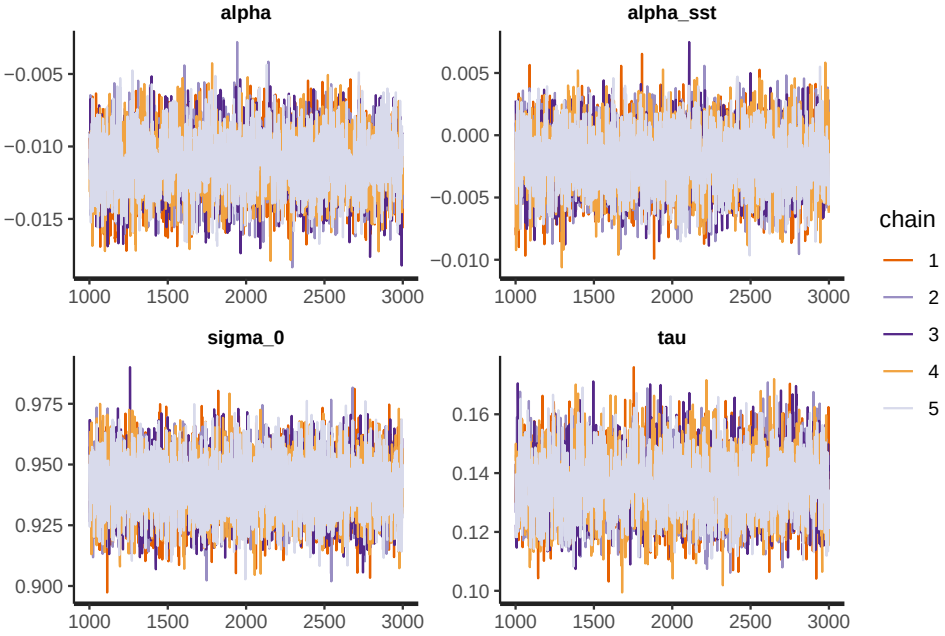
$$\epsilon_{i,t} \sim N(0, \sqrt{\delta_{it}}\tau) \tag{4}$$

Chris dug up a lot of temparature data, I summarised it and made α a function of temperature, $\alpha = \alpha_0 + \alpha_1 T_{it}$ with T_{it} being the standardized (i.e. mean 0, sd 1) water temperature for each site i in time period between $t - \delta_{it}$ and t. Finally, we observe X with error, leading to log-normal distributed observations with observation variability σ

$$Y_{i,t} = LogNormal(\log(X_{i,t}) - 0.5\sigma^2, \sigma) \tag{5}$$

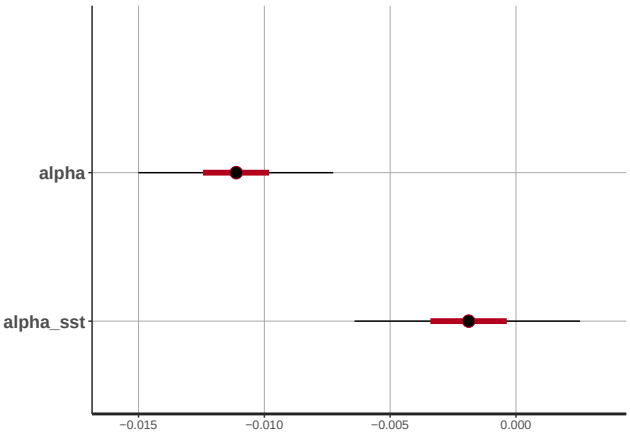
$$ADD - CENSORED - EQ - HERE \tag{6}$$

Ok. So what we care about are the estimated states (X_{it}), the decay parameters (α_0 and α_1), the process standard deviation τ and the observation standard deviation σ. I fit the model in stan. It worked fine but was a little fussy about starting conditions.

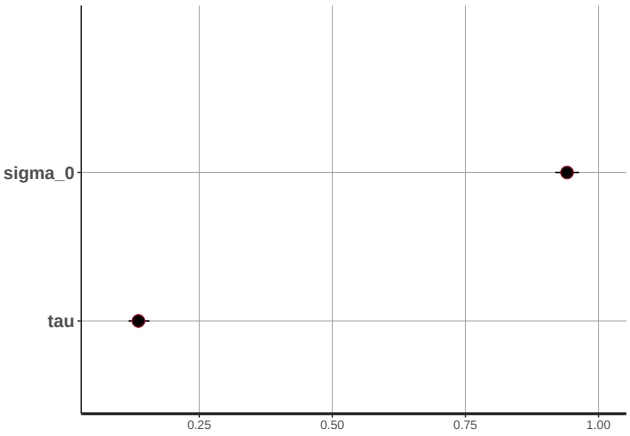



```
## ci_level: 0.5 (50% intervals)

## outer_level: 0.95 (95% intervals)
```




```
## ci_level: 0.5 (50% intervals)
## outer_level: 0.95 (95% intervals)
```



That seems reasonable enough. α is slightly negative, α_1("alpha_sst") is now slightly less than 0, the observation sd is big ("sigma_0") and the process sd is small but non-zero. Let's look at the fits. First we can look at the estimated states (Fig. 3). They look broadly ok (line goes through the points, more or less. Domoic acid generally, but not always, declines through time. Again, this is the estimate of the true domoic acid concentration. Next we can go look at what kinds of observations we would expect from this concentration of domoic acid.

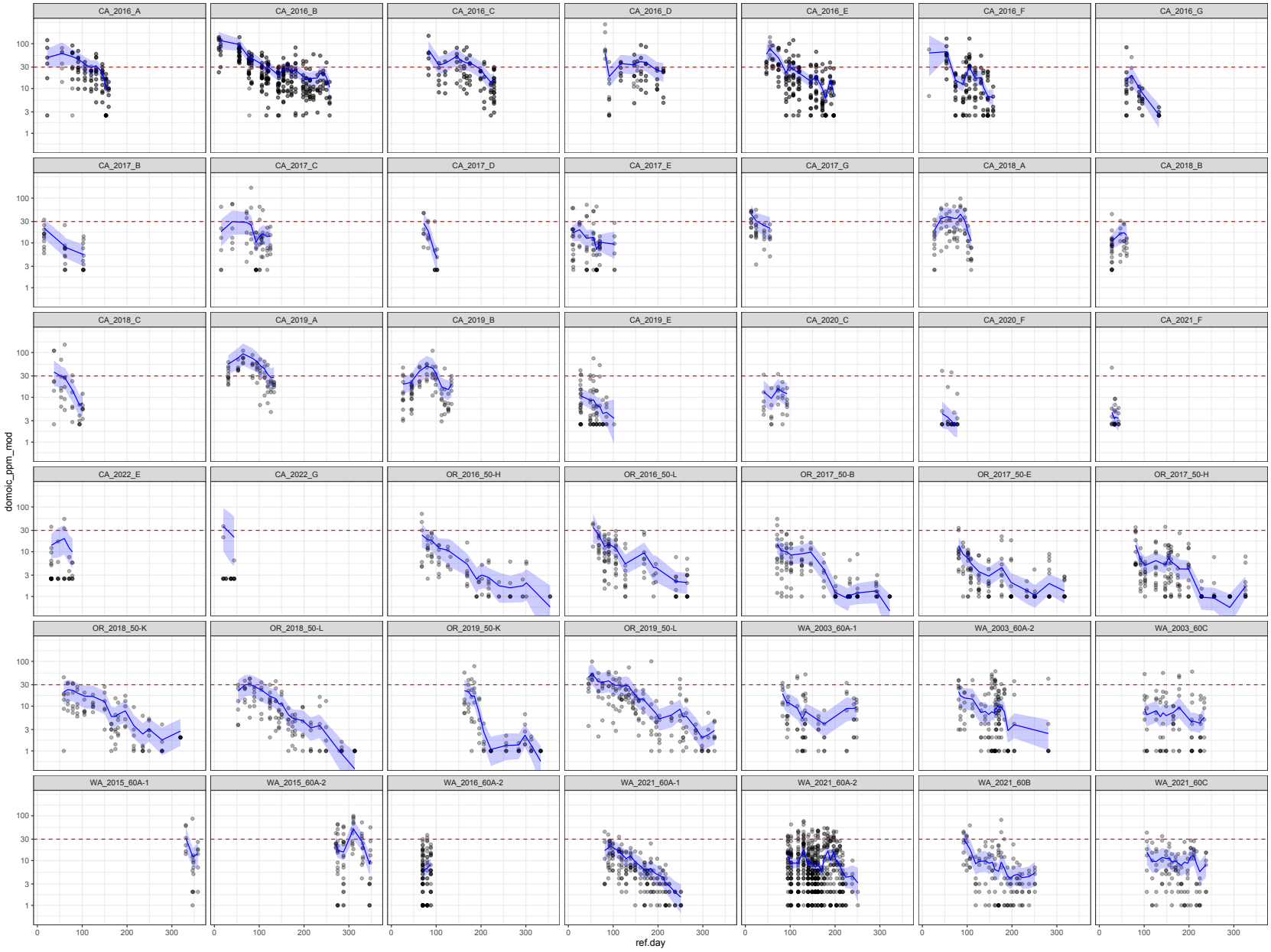


Figure 3: Blue is estimated state (Domoic ppm), Ribbon is 95 CI. Note log-scale on y-axis. Red line = 30 ppm domoic acid

We can plot posterior predictions, which are the distribution of new data that we would expect from our model. We'll plot it in red to differentiate it from the states. This should have a very similar mean to the estimated state, but because the observation sd (σ) is substantial, we expect the ribbon for this to be pretty large. And it is. Note that because of censoring, you never have observations below 2.5 ppm in California or 1.0 in Washington and Oregon (Fig. 4). Overall, the data generally falls within the ribbon. The data might be a little biased on the high side for some of the sites... but overall this doesn't seem too bad for noisy data.

This kind of plot is helpful because you can calculate the probability of observing a specific ppm conditioned on a true domoic acid ppm. The large measurement error means that even when true domoic acid ppm is pretty low (say < 10ppm), we will still occasionally find a sample that is > 30 ppm.

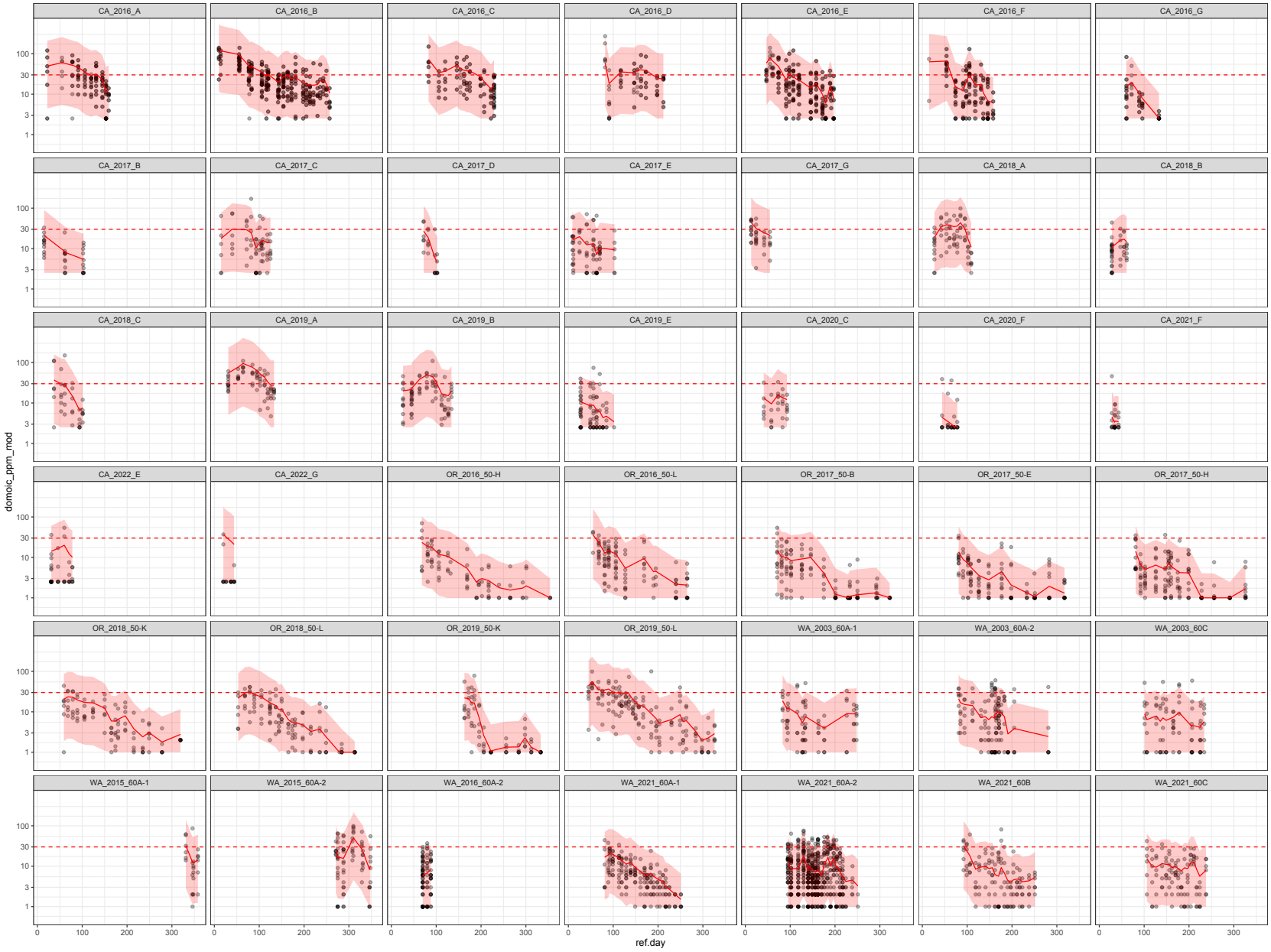


Figure 4: Red is mean estimate (Domoic ppm), Ribbon is 95 bound for predictions. Red line = 30 ppm domoic acid

And you can calculate, what was the highest domoic concentration for each of these time-series

So I feel ok about the model. There are some other bells and whistles that aren't too hard to add.

- Allow for different decay parameters (αs) spatially
  - Add other covariates?
- Make the observation variance state- or zone- specific
  - It's clear from the data that they are using different machines / protocols to measure domoic acid... so this is easy to motivate.

We can do a couple other things with it. For example, we can use the output to plot the estimated decay from known starting concentrations to a threshold level.

Or we can calculate the proportion of samples that would be expected to be less than a threshold of 30 ppm given a range of starting concentrations. (Fig. 7).

The figures can be prettied up a bunch... but that's where I got to this week.

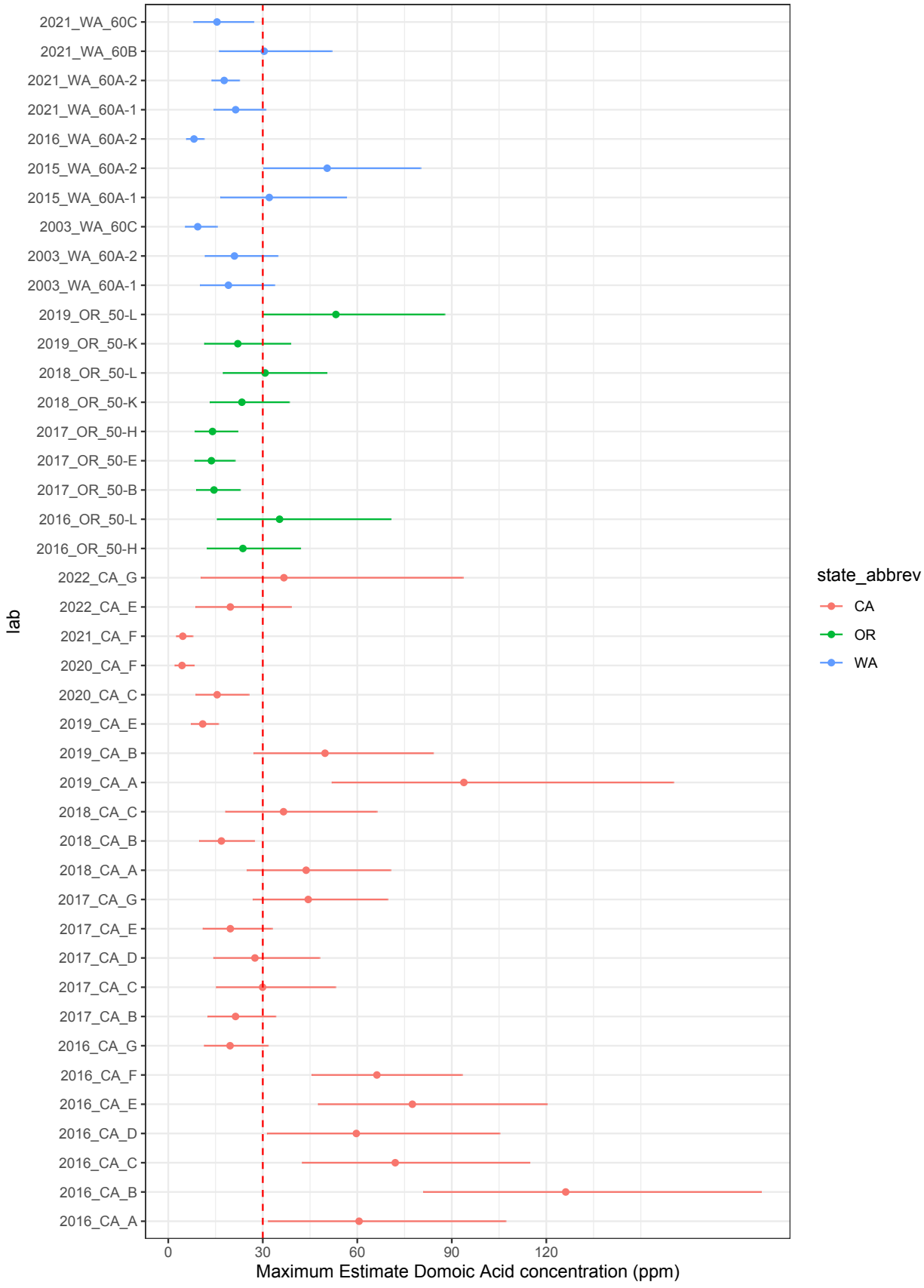


Figure 5: Mean estimate (95 CI) for time point with the highest domoic acid ppm. Red line = 30 ppm domoic acid

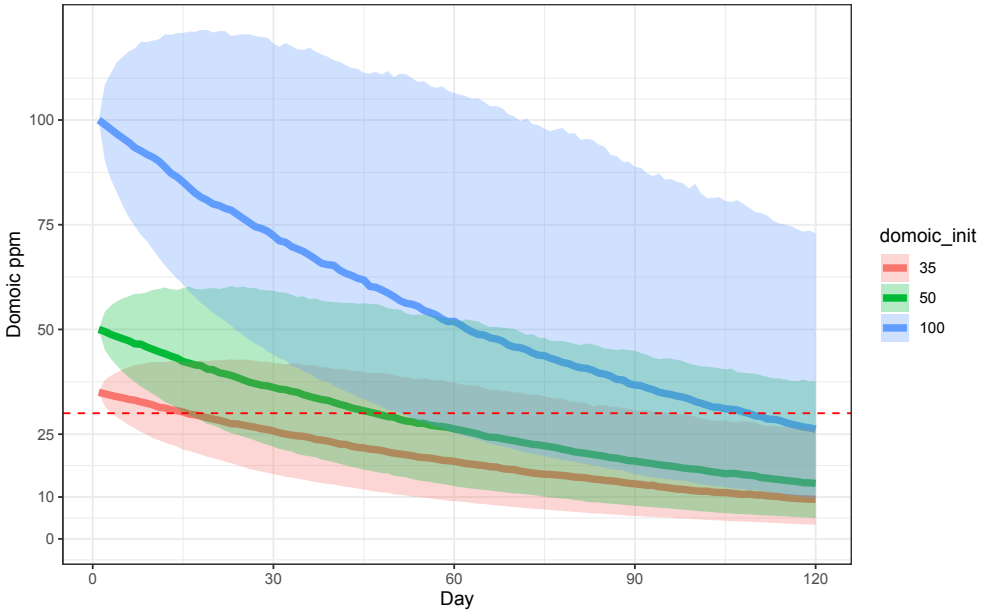


Figure 6: Median estimate and interquartile range decay from intial concentrations of 25, 50, and 100 ppm. Red line = 30 ppm domoic acid

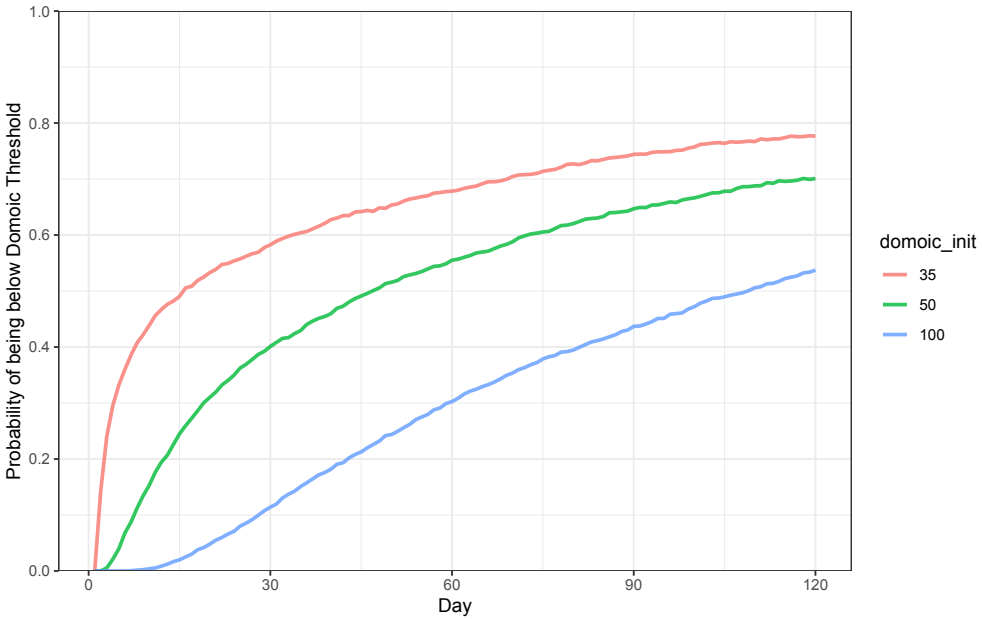


Figure 7: Proportion of 10,000 simulations in which the domoic concentration was lower than 30ppm at each day.